A new lower bound for the cap set capacity

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Abstract

Let $C = \lim_{n\to\infty} r_3(\mathbb{F}_3^n)^{1/n}$ be the cap set capacity, where $r_3(\mathbb{F}_3^n)$ is the largest size of a subset of $\mathbb{F}_3^n$ without three distinct points on a line. We show $C > 2.220334$; the explicit construction gives $C \geq 2.2203344$. This improves the bound 2.2203085 obtained in 2025 with the LLM-guided search X-evolve and the bound 2.2202337 of FunSearch (2023). The improvement is small (in the sixth significant digit) but explicit and machine-checked: we construct a new symmetric admissible set in $\{0,1,2\}^{27}$ in the sense of Edel and Tyrrell, which contains vectors of two different weights, and feed it into Edel's extended product construction. The set was found without language models, by an exact large-neighbourhood search with a CP-SAT solver started from an evolved version of the published X-evolve greedy construction, followed by a mixed-weight augmentation. The resulting statement — an explicit cap set in $\mathbb{F}_3^{648}$ of size larger than 2.220334^{648} — is verified in Lean 4 with Mathlib, including Edel's lemmas and the reduction from symmetric rows to admissible vectors; the large finite checks use `native_decide`.

1 Introduction

A *cap set* is a set $A \subseteq \mathbb{F}_3^n$ such that $x + y + z = 0$ with $x, y, z \in A$ forces $x = y = z$; equivalently A contains no three distinct collinear points. The sequence $r_3(\mathbb{F}_3^n)$ is supermultiplicative, so the *capacity* $C = \lim_n r_3(\mathbb{F}_3^n)^{1/n}$ exists, and Ellenberg and Gijswijt proved $C \leq 2.7552$ [3]. All known lower bounds come from explicit constructions: Calderbank and Fishburn [1] obtained 2.2101; Edel [2] 2.217389 with his *extended product* construction; Tyrrell [5] 2.218021 using SAT-solver searches for *admissible sets*; FunSearch [4] 2.2202337, with large language models evolving greedy priority programs that build symmetric admissible sets ($I(15, 10)$, and $A(24, 17)$ of size 237 984); and X-evolve [6] 2.2203085 with a symmetric admissible set $A(27, 19)$ of size 1 270 863. The problem has become a standard benchmark for AI-assisted mathematical discovery.

Theorem 1.1. *There is a cap set $A \subseteq \mathbb{F}_3^{648}$ with $|A| > 2.220334^{648}$; more precisely $|A|^{1/648} = 2.2203344\ldots$. Consequently $C \geq 2.2203344 > 2.2203085$.*

Ingredients. (i) A fast exact reimplementation of the FunSearch/X-evolve greedy, which makes thousands of evaluations affordable; (ii) parameter evolution of the X-evolve priority function; (iii) an exact large-neighbourhood search (LNS): neighbourhoods of 20–100 rows are removed and re-inserted optimally with a CP-SAT model; (iv) the observation that admissible sets need not have constant weight — in the final count a vector of weight $w + 1$ is worth $112/48$ vectors of weight w; (v) a Lean formalisation of the whole chain.

2 Extended products and admissible sets

A triple $(a,b,c) \in \{0,1,2\}^3$ is *good* if as a multiset it equals $\{0,1,2\}$, $\{0,0,1\}$ or $\{0,0,2\}$.

Definition 2.1 ([5, Def. 2.5])**.** $S \subseteq \{0,1,2\}^n$ is *admissible* if (i) for distinct $s, s' \in S$ there are i, j with $s_i = 0 \neq s'_i$ and $s_j \neq 0 = s'_j$, and (ii) for distinct $s, s', s'' \in S$ some coordinate (s_k, s'_k, s''_k) is good.

Definition 2.2 ([5, Def. 2.4])**.** Caps $A_0, A_1, A_2 \subseteq \mathbb{F}_3^d$ form an *extendable collection* if $x+y+z \neq 0$ whenever $x, y \in A_0$, $z \in A_1 \cup A_2$, and whenever $x \in A_0, y \in A_1, z \in A_2$.

For $s \in \{0,1,2\}^n$ let $s(A) = A_{s_1} \times \cdots \times A_{s_n} \subseteq \mathbb{F}_3^{dn}$ and $S(A) = \bigcup_{s \in S} s(A)$.

Lemma 2.3 (Edel [2, Lemma 10], [5, Lemmas 2.7, 2.11])**.** *If (A_0, A_1, A_2) is extendable and S is admissible, then $S(A)$ is a cap. If $|A_1| = |A_2|$, the blocks $s(A)$, $s \in S$, are pairwise disjoint and $|S(A)| = \sum_{s \in S} |A_0|^{n-|s|} |A_1|^{|s|}$, where $|s|$ is the number of nonzero entries.*

Sketch. Let $x \in s(A)$, $y \in s'(A)$, $z \in s''(A)$ with $x + y + z = 0$. If $s = s' = s''$ the blocks are caps and $x = y = z$. If exactly two labels coincide, say $s = s' \neq s''$, condition (i) gives a block with labels $(0, 0, \neq 0)$, contradicting extendability; if all three differ, (ii) gives a block with labels $\{0, 0, \neq 0\}$ or $\{0, 1, 2\}$, again a contradiction. Disjointness follows from (i) and $A_0 \cap (A_1 \cup A_2) = \emptyset$ (extendability with $x = y = z$, using $3x = 0$). Constant weight is not needed for the count. □

Lemma 2.4 (Edel; [5, Lemmas 2.9, 2.12])**.** *Let $\tilde{I}(m, m-1)$ consist of the m vectors with exactly one zero, entries before the zero equal to 1 and after it equal to 2. If (A_0, A_1, A_2) is extendable, so is $(\tilde{I}(m, m-1)(A), A_1^m, A_2^m)$.*

With the Calderbank–Fishburn collection in $\mathbb{F}_3^6$ of sizes $(12, 112, 112)$ [5, Lemma 2.14] and $m = 4$ we obtain an extendable collection in $\mathbb{F}_3^{24}$ with sizes $b_0 = 4 \cdot 12 \cdot 112^3$ and $b_1 = b_2 = 112^4$. An admissible $S \subseteq \{0,1,2\}^{27}$ then gives a cap in $\mathbb{F}_3^{648}$ of size

$$M(S) = \sum_{s \in S} b_0^{27-|s|}\, b_1^{|s|}. \tag{2.1}$$

For constant weight w this is $|S|\, b_0^{27-w} b_1^w$, the formula used in [4, 6]; $m = 4$ is the best choice for $w = 19$.

3 Symmetric admissible sets

Following [4] we split the 27 coordinates into 9 groups of 3 and use the seven block types $000, 001, 002, 012, 021, 111, 222$, numbered $0, \ldots, 6$, of weights $0, 1, 1, 2, 2, 3, 3$. A *row* $t \in \{0, \ldots, 6\}^9$ stands for the set $E(t)$ of all vectors whose i-th block is a cyclic rotation of the block of type t_i; $|E(t)| = 3^{k(t)}$, where $k(t)$ is the number of groups of weight 1 or 2. The *profile* of t is its vector of block weights. A type triple (a, b, c) is *always good* if every choice of rotations of the three blocks has a good coordinate (exactly 103 of the 343 ordered type triples are not).

Theorem 3.1 (type-level criterion)**.** *Let R be a set of rows such that* (P) *the profiles of distinct rows are incomparable in the componentwise order, and* (T) *any three distinct rows have a group in which their types form an always-good triple. Then $S = \bigcup_{t \in R} E(t)$ is admissible, the sets $E(t)$ are pairwise disjoint, and $|S| = \sum_{t \in R} 3^{k(t)}$.*

Proof. We use four finite facts about single blocks, verified by exhaustion: (a) two different rotations of one type have a coordinate where the first is 0 and the second is not; (b) three rotations of one type, not all equal, have a good coordinate; (c) if $\operatorname{wt} b < \operatorname{wt} a$, any rotation of a has a nonzero coordinate where any rotation of b is zero; (d) if $\operatorname{wt} a < \operatorname{wt} b$, then two (possibly equal) rotations of a and any rotation of b have a good coordinate. Let $v \neq v'$ in S come from rows t, t'. If $t = t'$, the vectors differ in some group and (a) gives condition (i). If $t \neq t'$, (P) gives groups i, j with $\operatorname{wt} t_i > \operatorname{wt} t'_i$ and $\operatorname{wt} t_j < \operatorname{wt} t'_j$, and (c) gives (i). Now let $u, v, w \in S$ be distinct, from rows t, t', t''. If $t = t' = t''$, use (b) in a group where u, v, w are not all equal. If exactly two rows coincide, (P) yields a group where the third row is heavier, and (d) applies (the good property is symmetric). If the rows are distinct, (T) applies. Vectors of $E(t) \cap E(t')$ would have equal profiles, contradicting (P). □

For constant weight, (P) says only that distinct rows have distinct profiles (the setting of [4, 6]). Allowing different weights is new here; by (2.1) a vector of weight 20 counts $b_1/b_0 = 112/48$ times as much as one of weight 19.

Remark 3.2 (a tempting but invalid shortcut). Tyrrell's "meta-extension" [5, §3] combines admissible sets S_0, S_1, S_2. With $S_2 = S_1$ and S_0 of weight $w - 1$ it looks very profitable, but its condition (3) must also hold for $y = z \in S_1$, which forces $\operatorname{supp} x \not\subseteq \operatorname{supp} y$ for $x \in S_0$, i.e. the same antichain condition as in Theorem 3.1; the meta product is then worse than using $S_0 \cup S_1$ directly. With S_2 the $1 \leftrightarrow 2$ swap of S_1 condition (3) leaves almost no candidates of weight $w - 1$.

4 The search

Fast greedy. FunSearch and X-evolve build sets greedily: rows are sorted by a priority function and added when compatible. We reimplemented the greedy in C with a bitmap over the 7^9 type codes that marks rows blocked by a pair of chosen rows (the blocked rows of a pair (x, y) form a box $\prod_i B(x_i, y_i)$ intersected with the weight condition). It reproduces FunSearch's $A(24, 17)$ of size 237 984 in 1.5 s and X-evolve's $A(27, 19)$ of size 1 270 863 in about 15 s from the published priority program (scores rounded to single precision, as in X-evolve's C++ helper).

Parameter evolution. A (1+1) evolution strategy over the coefficients of the X-evolve priority plus pairwise correction terms gave 1 272 537.

Large-neighbourhood search. From there we repeatedly remove 20–100 rows — either rows with nearby profiles, or a vertex cover of the pairs of chosen rows that block a promising unused row (preferably of high value 3^k) — list all rows that become insertable, and solve the re-insertion problem exactly with OR-tools CP-SAT: binary variables for candidate rows, at most one row per profile, pairwise conflicts induced by bad triples with a fixed row, and triple conflicts among candidates added lazily; the move is accepted if the total does not decrease. Since the X-evolve set is almost invariant under cyclic shifts of the 9 groups, every improving move is also tried in its 8 shifted versions. About 10^4 neighbourhoods were solved in total (roughly 20 CPU hours over two days).

Mixed weights. Finally rows of weight 18 and 20 whose profiles are incomparable with all chosen profiles and which create no bad triple are added greedily, followed by a short iterated local search on the mixed objective (2.1).

5 The new set

construction	$\lvert S\rvert$ (or weighted)	bound on C	source
$A(24,17)$	237 984	2.2202337	FunSearch [4]
$A(27,19)$	1 270 863	2.2203085	X-evolve [6]
$A(27,19)$ + weight 20	1 271 409 (weighted)	2.2203100	this paper, Sec. 3
LNS + mixed weights	1 280 534 (weighted)	2.2203344	this paper

The final row set has 1768 rows; its expansion has 1 280 424 vectors of weight 19 and 42 of weight 20 and 28 of weight 18. The weighted size is $\sum_s (b_1/b_0)^{|s|-19}$. The rows are listed in `data/rows.txt` (one row of 9 types per line) and, base-7 encoded, in the Lean file.

6 Verification

Independent checks. A Python script recomputes the always-good table from scratch (by brute force over rotations), checks (P) and (T) with bitsets, and recomputes (2.1); the C search code re-inserts every row through its own conflict structures; and Theorem 3.1 was compared with the definition on expanded vectors for all pairs of rows and 3000 random row sets with at most 3 groups.

Lean. The file `lean/Main.lean` (Lean 4.35.0, Mathlib) proves

```
theorem main_cap_648 : exists A : Finset (Fin 648 -> ZMod 3),
    IsCap A /\ (NUM : Real) ^ 648 < (A.card : Real) * (DEN : Real) ^ 648

theorem main_capacity (k : Nat) (hk : 1 <= k) :
    exists B : Finset (Fin k -> Fin 9 x Fin 3 -> Fin 4 -> Fin 6 -> ZMod 3),
      IsCap B /\ (NUM : Real) ^ (648 * k)
                  < (B.card : Real) * (DEN : Real) ^ (648 * k)
```

with NUM/DEN = 2220334/1000000, where `IsCap A` means $\forall x,y,z \in A,\ x+y+z=0 \Rightarrow x=y \wedge y=z$. Lemmas 2.3 and 2.4, the count (2.1), Theorem 3.1 and the soundness of a bitset checker for (P) and (T) are proved in Lean. The checker runs on the explicit rows, and the facts about the explicit Calderbank–Fishburn collection are evaluated, by `native_decide` (about two minutes); hence the trusted base includes Lean's compiler and interpreter. The final inequality between integers with about 3900 digits is checked by the kernel. `#print axioms` lists `propext`, `Classical.choice`, `Quot.sound` and the auxiliary axioms produced by `native_decide`.

7 Limitations and outlook

The gain over X-evolve is in the sixth significant digit: it shows that exact neighbourhood search improves greedy constructions of symmetric admissible sets and that mixed weights are available, but the gap to the limit $124^{1/6} \approx 2.2331$ of the method [5] (if all admissible sets $I(m,w)$ existed) remains. The last steps — monotonicity of r_3 in n and the existence of C — are standard and not formalised; Lean gives caps in all dimensions $648k$. We did not try other base collections beyond a small SAT search in $\mathbb{F}_3^4$, nor other symmetry groups.

References

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[5] F. Tyrrell, New lower bounds for cap sets, *Discrete Analysis* 2023:20; arXiv:2209.10045.

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